

Zhaokai Wang (王肇凯)

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HIGHLIGHTS

- **Core Contributor of InternVL Series:** Key member of open-source models InternVL3.5 (10K+ Github stars, 1200+ citations), InternVL-U and Mono-InternVL (100 citations) at Shanghai AI Laboratory.
- **Prolific Publications:** Published 10+ first-author or co-first-author papers in top-tier conferences/journals including TPAMI, NeurIPS, ICML, CVPR, ICCV.
- **Research Impact:** Accumulated 3000+ citations. Receive Best Paper Runner-up at NeurIPS 2025, Best Paper Award at ACM Multimedia 2021, and three workshop best paper awards.
- **Research Interest:** Vision-language models, coding agents, music generation.

EDUCATION

Shanghai Jiao Tong University	09/2022 - 06/2027 (Expected)
Ph.D. Student (Joint Program with Shanghai AI Laboratory)	Advisor: Prof. Jifeng Dai
Peking University	09/2019 - 06/2022
Double Bachelor Degree in Economics	
Beihang University	09/2018 - 06/2022
B.Eng in Computer Science	Overall GPA: 3.80

EXPERIENCE

Naive AI	Shanghai
Collaborated with Prof. Jifeng Dai, working on coding agents	07/2026 - Now
OpenGVLab (InternVL Team), Shanghai AI Laboratory	Shanghai
Collaborated with Prof. Jifeng Dai and Dr. Xizhou Zhu, working on multimodal learning	11/2022 - 06/2026
Fundamental Vision Group, SenseTime	Beijing
Collaborated with Prof. Jifeng Dai and Dr. Xizhou Zhu, working on multimodal learning	02/2022 - 10/2022
Sea AI Lab	Beijing
Collaborated with Prof. Shuicheng Yan and Dr. Jibin Wu, working on music generation	08/2021 - 02/2022
CoLab, Institute of Artificial Intelligence at Beihang University	Beijing
Collaborated with Prof. Si Liu, working on computer vision	08/2019 - 06/2022

AWARDS AND HONORS

- Honors
 - Tencent Hunyuan Scholar (¥100,000 scholarship)** 2026
 - Outstanding Graduate, Beihang University 2022
- Main Conference
 - Best Paper Runner Up Award**, NeurIPS 2025 2025
 - Best Paper Award**, ACM Multimedia 2021 2021
- Workshops and Competitions
 - Best Paper Award**, ICML AI4MATH Workshop 2025 2025
 - Best Paper Award**, IJCAI MKLM Workshop 2025 2025
 - Best Paper Award**, KDD Urban Computing Workshop (UrbComp) 2024 2024
 - First Place, CVPR 2020 TextCaps Challenge 2020

PUBLICATIONS

MLLM, Unified Models and Image Generation

1. Parameter-Inverted Image Pyramid Networks for Visual Perception and Multimodal Understanding
Zhaokai Wang, Xizhou Zhu, Xue Yang, Gen Luo, Hao Li, Changyao Tian, Wenhan Dou, Junqi Ge, Lewei Lu, Yu Qiao, Jifeng Dai
TPAMI 2025
2. GenExam: A Multidisciplinary Text-to-Image Exam
Zhaokai Wang, Penghao Yin, Xiangyu Zhao, Changyao Tian, Yu Qiao, Wenhai Wang, Jifeng Dai, Gen Luo
ICML 2026

3. Mono-InternVL: Pushing the Boundaries of Monolithic Multimodal Large Language Models with Endogenous Visual Pre-training
Gen Luo*, Xue Yang*, Wenhan Dou*, Zhaokai Wang*, Jiawen Liu, Jifeng Dai, Yu Qiao, Xizhou Zhu
CVPR 2025
4. Confidence-aware Non-repetitive Multimodal Transformers for TextCaps
Zhaokai Wang, Renda Bao, Qi Wu, Si Liu
AAAI 2021
5. DisciplineGen-1M: A Large-Scale Dataset for Multidisciplinary Visual Generation and Editing
Zhaokai Wang, Mingxin Liu, Zirun Zhu, Ziqian Fan, Yiguo He, Mohan Zhang, Leyao Gu, Xiangyu Zhao, Ning Liao, Shaofeng Zhang, Xuanhe Zhou, Zhihang Zhong, Junchi Yan, Xue Yang
Preprint
6. InternVL3.5: Advancing Open-Source Multimodal Models in Versatility, Reasoning, and Efficiency
Weiyun Wang*, Zhangwei Gao*, Lixin Gu*, Hengjun Pu*, Long Cui*, Xingguang Wei*, Zhaoyang Liu*, Linglin Jing*, Shenglong Ye*, Jie Shao*, Zhaokai Wang*, Zhe Chen*, Hongjie Zhang, Ganlin Yang, Haomin Wang, Qi Wei, Jinhui Yin, Wenhao Li, Erfei Cui, Guanzhou Chen, Zichen Ding, Changyao Tian, Zhenyu Wu, Jingjing Xie, Zehao Li, Bowen Yang, Yuchen Duan, Xuehui Wang, Zhi Hou, Haoran Hao, Tianyi Zhang, Songze Li, Xiangyu Zhao, Haodong Duan, Nianchen Deng, Bin Fu, Yinan He, Yi Wang, Conghui He, Botian Shi, Junjun He, Yingdong Xiong, Han Lv, Lijun Wu, Wenqi Shao, Kaipeng Zhang, Huipeng Deng, Biqing Qi, Jiaye Ge, Qipeng Guo, Wenwei Zhang, Songyang Zhang, Maosong Cao, Junyao Lin, Kexian Tang, Jianfei Gao, Haiyan Huang, Yuzhe Gu, Chengqi Lyu, Huanze Tang, Rui Wang, Haijun Lv, Wanli Ouyang, Limin Wang, Min Dou, Xizhou Zhu, Tong Lu, Dahua Lin, Jifeng Dai, Weijie Su, Bowen Zhou, Kai Chen, Yu Qiao, Wenhao Wang, Gen Luo
Technical Report
7. InternVL-U: Democratizing Unified Multimodal Models for Understanding, Reasoning, Generation and Editing
Changyao Tian*, Danni Yang*, Guanzhou Chen*, Erfei Cui*, Zhaokai Wang*, Yuchen Duan*, Penghao Yin*, Sitao Chen, Ganlin Yang, Mingxin Liu, Zirun Zhu, Ziqian Fan, Leyao Gu, Haomin Wang, Qi Wei, Jinhui Yin, Xue Yang, Zhihang Zhong, Qi Qin, Yi Xin, Bin Fu, Yihao Liu, Jiaye Ge, Qipeng Guo, Gen Luo, Hongsheng Li, Yu Qiao, Kai Chen, Hongjie Zhang
Technical Report
8. GRADE: Benchmarking Discipline-Informed Reasoning in Image Editing
Mingxin Liu*, Ziqian Fan*, Zhaokai Wang* (*Project Lead*), Leyao Gu*, Zirun Zhu*, Yiguo He, Yuchen Yang, Changyao Tian, Xiangyu Zhao, Ning Liao, Shaofeng Zhang, Qibing Ren, Zhihang Zhong, Xuanhe Zhou, Junchi Yan, Xue Yang
ECCV 2026 Oral
9. Sparkle: Mastering Basic Spatial Capabilities in Vision Language Models Elicits Generalization to Composite Spatial Reasoning
Yihong Tang*, Ao Qu*, Zhaokai Wang*, Dingyi Zhuang*, Zhaofeng Wu, Wei Ma, Shenhao Wang, Yunhan Zheng, Zhan Zhao, Jinhua Zhao
EMNLP 2025 Findings & IJCAI 2025 MKLM Workshop Best Paper Award
10. Mono-InternVL-1.5: Towards Cheaper and Faster Monolithic Multimodal Large Language Models
Gen Luo, Wenhan Dou, Wenhao Li, Zhaokai Wang, Xue Yang, Changyao Tian, Hao Li, Weiyun Wang, Wenhao Wang, Xizhou Zhu, Yu Qiao, Jifeng Dai
Technical Report
11. SynerGen-VL: Towards Synergistic Image Understanding and Generation with Vision Experts and Token Folding
Hao Li, Changyao Tian, Jie Shao, Xizhou Zhu, Zhaokai Wang, Jinguo Zhu, Wenhan Dou, Xiaogang Wang, Hongsheng Li, Lewei Lu, Jifeng Dai
CVPR 2025

LLM and Agents

12. WorldCupArena: Fine-Grained Evaluation of Language Models and Deep-Research Agents on Football Forecasting
Zhaokai Wang, Tianlin Gui, Jiayuan Rao, Shangzhe Di, Yihong Tang, Dingli Liang
Preprint
13. Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?
Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, Gao Huang
NeurIPS 2025 Best Paper Runner Up Award
14. Auto MC-Reward: Automated Dense Reward Design with Large Language Models for Minecraft
Hao Li*, Xue Yang*, Zhaokai Wang*, Xizhou Zhu, Jie Zhou, Yu Qiao, Xiaogang Wang, Hongsheng Li, Lewei Lu, Jifeng Dai
CVPR 2024

15. ITINERA: Integrating Spatial Optimization with Large Language Models for Open-domain Urban Itinerary Planning
Yihong Tang*, Zhaokai Wang*, Ao Qu*, Yihao Yan*, Zhaofeng Wu, Dingyi Zhuang, Jushi Kai, Kebin Hou, Xiaotong Guo, Jinhua Zhao, Zhan Zhao, Wei Ma
EMNLP 2024 Industry Track & KDD 2024 UrbComp Workshop Best Paper Award
16. OS Agents: A Survey on MLLM-based Agents for Computer, Phone and Browser Use
Xueyu Hu, Tao Xiong, Biao Yi, Zishu Wei, Ruixuan Xiao, Yurun Chen, Jiasheng Ye, Meiling Tao, Xiangxin Zhou, Ziyu Zhao, Yuhuai Li, Shengze Xu, Shawn Wang, Xinchun Xu, Shuofei Qiao, Zhaokai Wang, Kun Kuang, Tieyong Zeng, Liang Wang, Jiwei Li, Yuchen Eleanor Jiang, Wangchunshu Zhou, Guoyin Wang, Keting Yin, Zhou Zhao, Hongxia Yang, Fan Wu, Shengyu Zhang, Fei Wu
ACL 2025 Oral

Music Generation

17. Vision-to-Music Generation: A Survey
Zhaokai Wang, Chenxi Bao, Le Zhuo, Jingrui Han, Yang Yue, Yihong Tang, Victor Shea-Jay Huang, Yue Liao
ISMIR 2025
18. Video Background Music Generation: Dataset, Method and Evaluation
Le Zhuo*, Zhaokai Wang*, Baisen Wang*, Yue Liao, Chenxi Bao, Stanley Peng, Songhao Han, Aixi Zhang, Fei Fang, Si Liu
ICCV 2023
19. Video Background Music Generation with Controllable Music Transformer
Shangzhe Di, Zeren Jiang, Si Liu, Zhaokai Wang, Leyan Zhu, Zexin He, Hongming Liu, Shuicheng Yan
ACM Multimedia 2021 Best Paper Award
20. Multimodal Music Generation with Explicit Bridges and Retrieval Augmentation
Baisen Wang, Le Zhuo, Zhaokai Wang, Chenxi Bao, Chengjing Wu, Xuecheng Nie, Jiao Dai, Jizhong Han, Yue Liao, Si Liu
ISMIR 2025 LLM4Music Workshop

ADDITIONAL INFORMATION

- **Conference & Journal Reviewer:** TPAMI, NeurIPS, ICLR, ICML, ICCV, ECCV, CVPR, EMNLP, AAAI, ACM Computing Survey.
- **Teaching Assistant:** Fundamentals of Computers (2021), Software Engineering (2022).
- **Programming languages:** Python, C/C++, Java, JavaScript.
- **English:** TOEFL 111 (S26) , GRE 327+4.0, CET-4 669, CET-6 612.